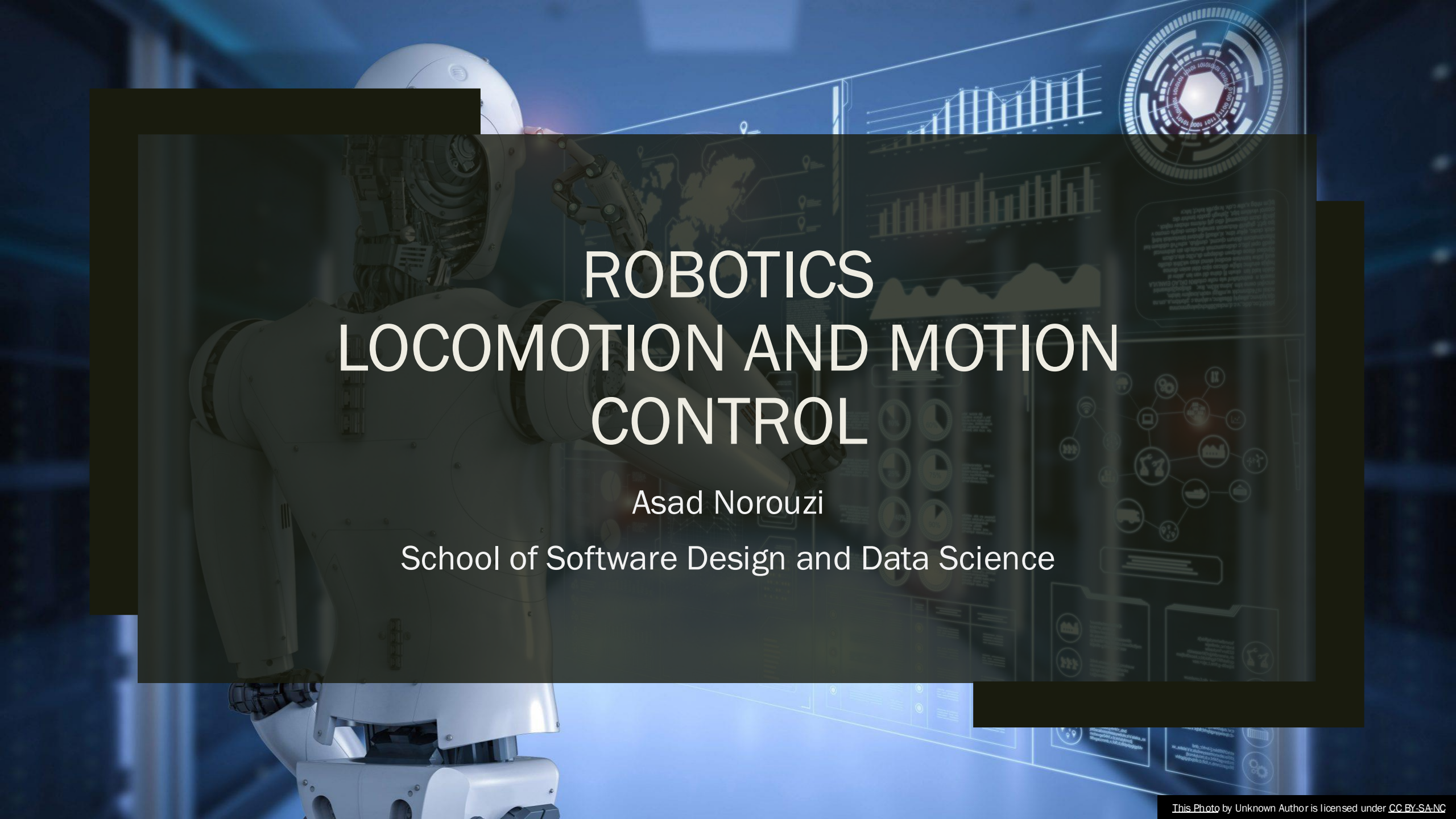


A white humanoid robot is positioned on the left side of the frame, facing right. It has a sleek, modern design with visible joints and a helmet-like head. The background is a dark blue, futuristic environment with glowing digital elements. On the right, there are several data visualizations: a bar chart at the top, a line graph below it, and a circular gauge or radar chart at the top right. A world map is faintly visible in the center background. The overall aesthetic is high-tech and digital.

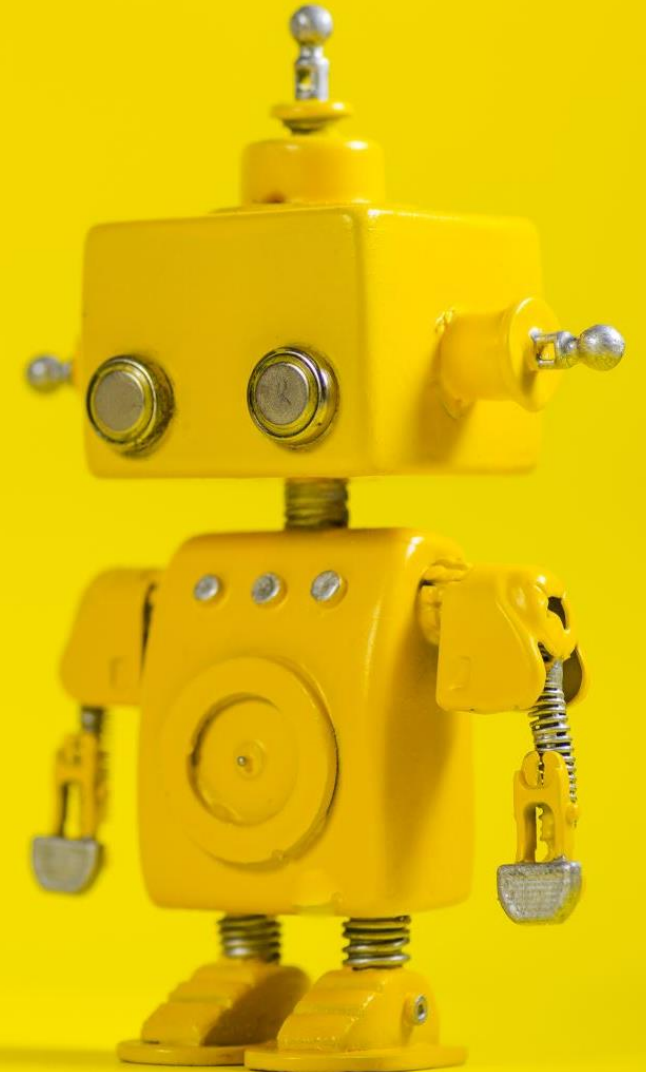
ROBOTICS LOCOMOTION AND MOTION CONTROL

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Agenda

- Introduction to Locomotion
- Types of Locomotion
- Wheeled Locomotion
- Legged Locomotion
- Tracked Locomotion
- Aerial Locomotion
- Aquatic Locomotion
- Motion Control Overview
- Kinematics in Motion Control
- Dynamics in Motion Control
- Control Strategies
- Path Planning and Navigation
- Challenges in Motion Control
- Summary
- Q&A





Introduction to Locomotion

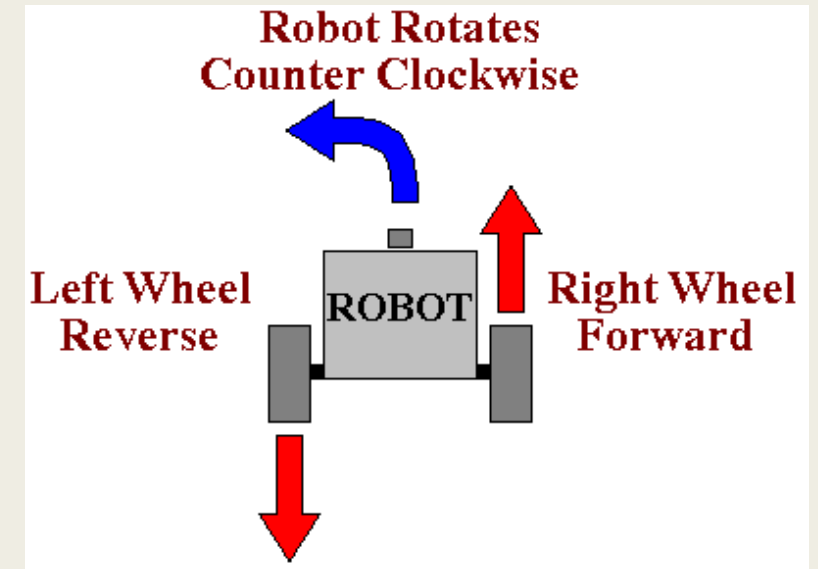
- Locomotion refers to the mechanisms that enable robots to move from one place to another.
- Essential for applications where robots need to navigate their environment, such as exploration, search and rescue, and service robots.

Types of Locomotion

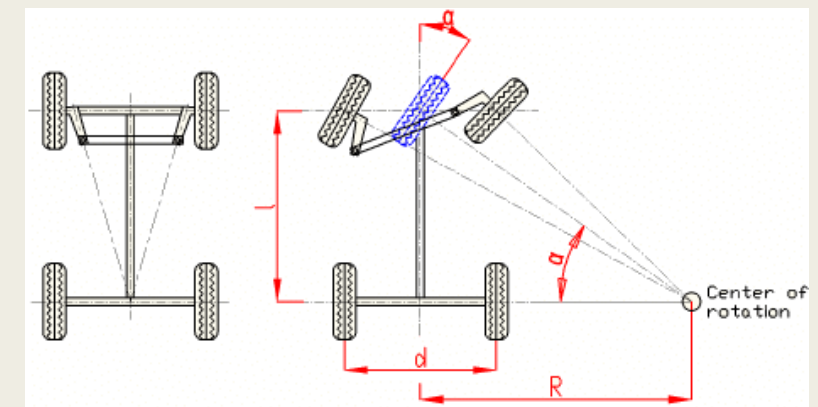
- **Wheeled Locomotion:**
 - *Uses wheels for movement, providing high speed and efficiency on flat surfaces.*
 - *Differential drive robots, car-like robots.*
- **Legged Locomotion:**
 - *Uses legs to move, allowing traversal over rough and uneven terrain.*
 - *Bipeds, quadrupeds, hexapods.*
- **Tracked Locomotion:**
 - *Uses continuous tracks, providing stability and the ability to navigate over obstacles.*
 - *Tank-like robots.*
- **Aerial Locomotion:**
 - *Uses propellers or wings to achieve flight.*
 - *Drones, UAVs.*
- **Aquatic Locomotion:**
 - *Uses fins, flippers, or jets to move in water.*
 - *Underwater robots, autonomous submarines.*

Wheeled Locomotion

- **Mechanisms:**
 - **Differential Drive:** Uses two independently driven wheels; allows precise control of direction and speed.
 - **Ackermann Steering:** Similar to car steering, where front wheels turn to steer.
- **Advantages:** High speed, energy efficiency, and simplicity in control and implementation.
- **Applications:** Indoor robots, delivery robots, automated guided vehicles (AGVs).



Differential Drive



Ackermann Steering

Legged Locomotion

- **Mechanisms:**
 - *Bipeds:* Two-legged robots mimicking human walking.
 - *Quadrupeds:* Four-legged robots providing stability and robustness.
 - *Hexapods:* Six-legged robots offering high stability and flexibility.
- **Advantages:** Ability to navigate rough and uneven terrain, greater adaptability.
- **Challenges:** Complex control and stability issues, higher energy consumption.
- **Applications:** Search and rescue, exploration, robotic pets.



Tracked Locomotion

- **Mechanisms:**
 - ***Continuous Tracks:** Provide continuous contact with the ground, allowing traversal over obstacles.*
 - ***Track Steering:** Differential steering similar to wheeled robots, by varying the speed of tracks.*
- **Advantages:** Stability on rough terrain, ability to climb obstacles, high traction.
- **Challenges:** Slower speed compared to wheels, complex track maintenance.
- **Applications:** Military robots, construction robots, exploratory robots.



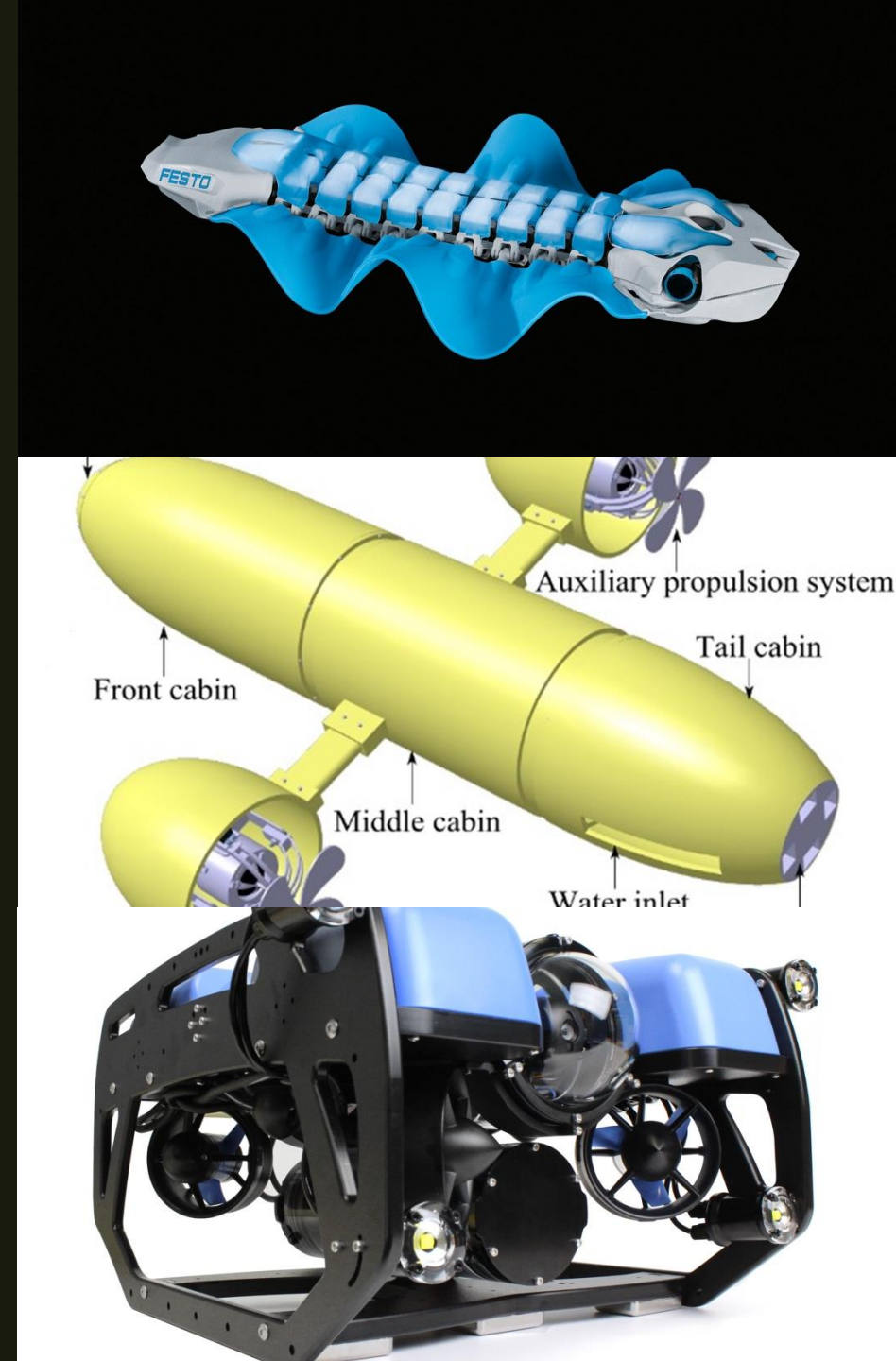
Aerial Locomotion

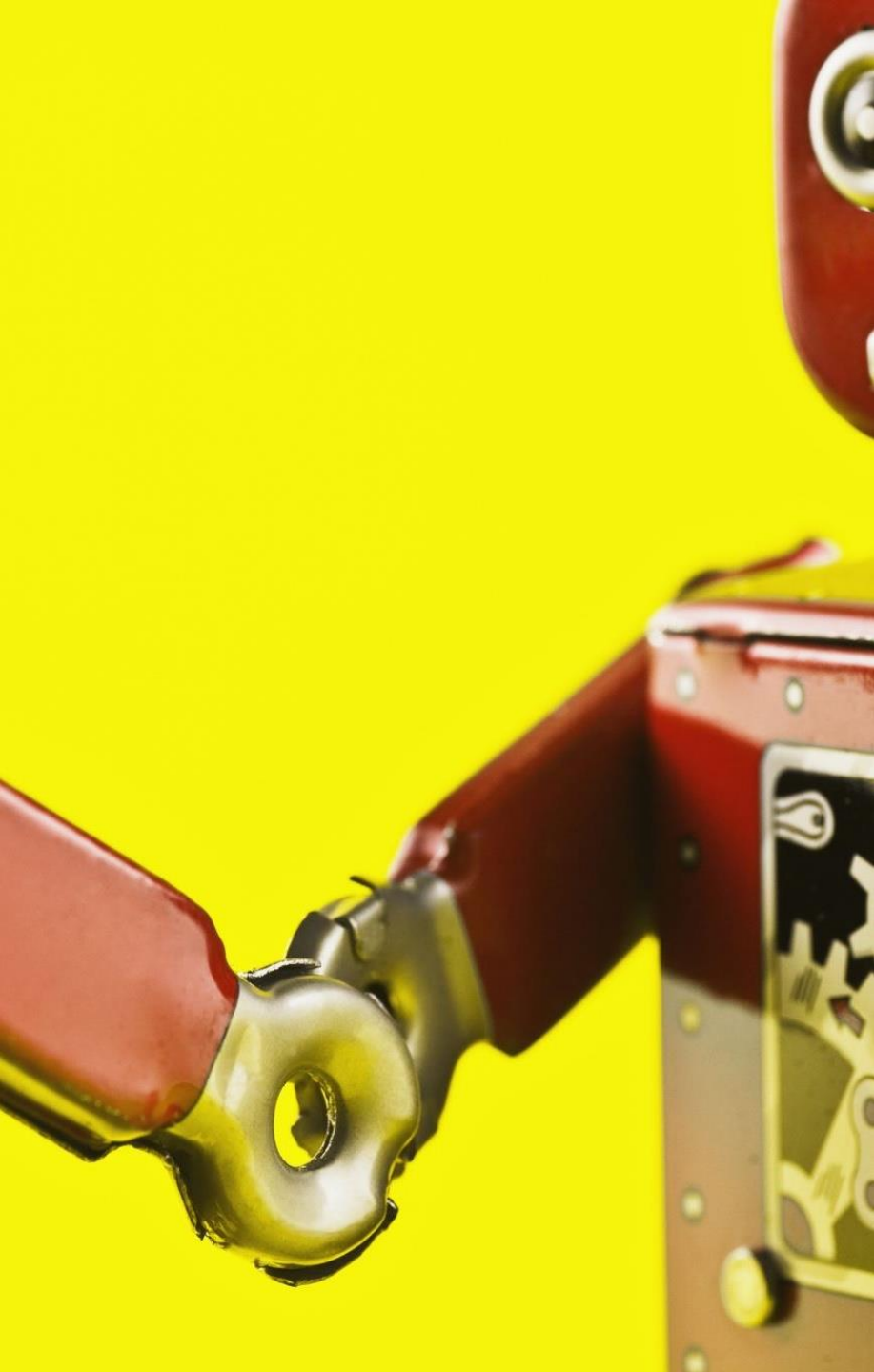
- **Mechanisms:**
 - ***Multirotors:** Use multiple rotors for lift and control.*
 - ***Fixed-Wing Aircraft:** Use wings for lift and propellers for thrust.*
- **Advantages:** Ability to cover large areas quickly, access to hard-to-reach places.
- **Challenges:** Limited flight time due to battery life, complex control in windy conditions.
- **Applications:** Aerial surveillance, delivery drones, environmental monitoring.



Aquatic Locomotion

- **Mechanisms:**
 - *Propeller-Based:* Uses propellers for thrust.
 - *Jet Propulsion:* Expels water to generate thrust.
 - *Undulatory Motion:* Mimics the movement of fish.
- **Advantages:** Efficient movement in water, ability to explore underwater environments.
- **Challenges:** Waterproofing and pressure resistance, communication challenges underwater.
- **Applications:** Underwater exploration, marine biology research, underwater maintenance.



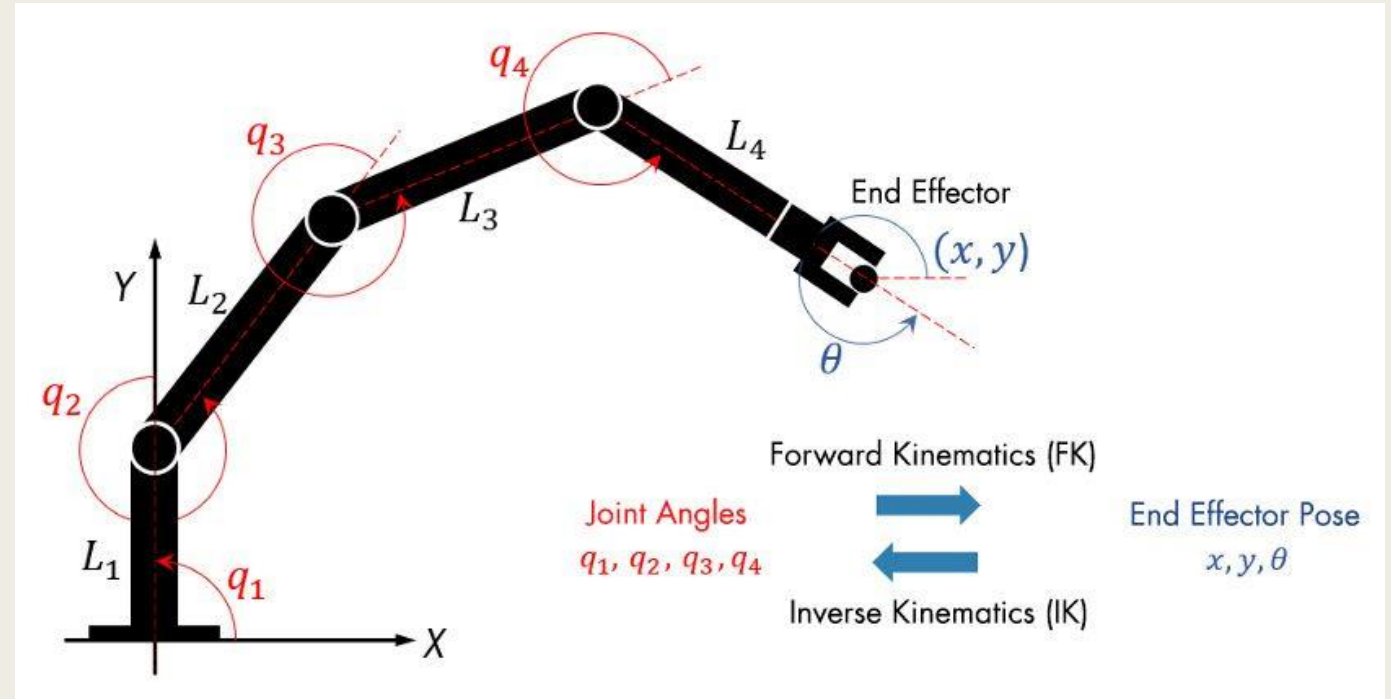


Motion Control Overview

- Motion control refers to the techniques and algorithms used to control the movement of robots.
- Ensures precise and efficient movement, allowing robots to navigate and perform tasks effectively.

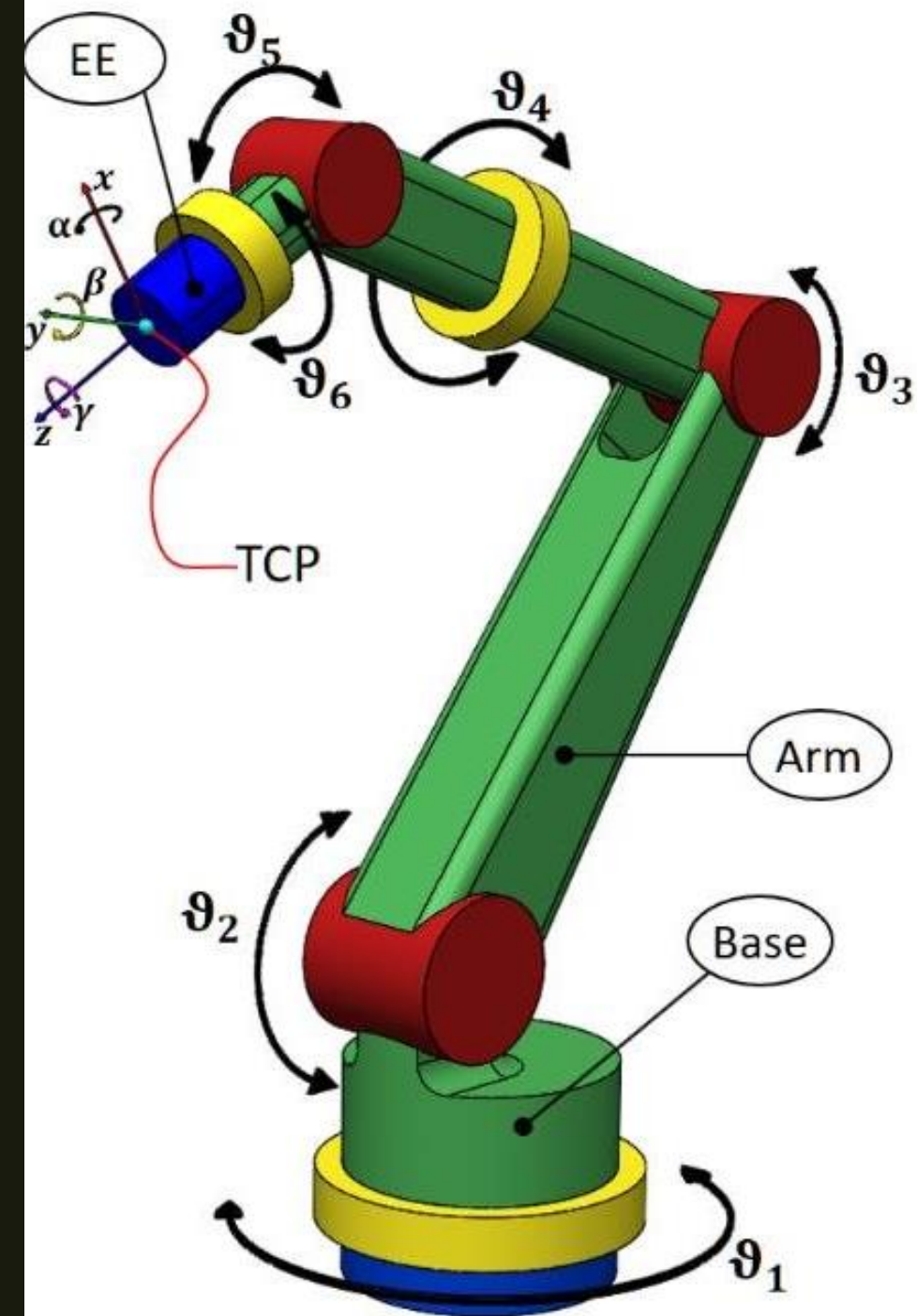
Kinematics in Motion Control

- Kinematics studies the motion of robots without considering the forces that cause the motion.
- Types:
 - **Forward Kinematics:**
Computes the position and orientation of the robot's end-effector given its joint parameters.
 - **Inverse Kinematics:**
Computes the joint parameters needed to achieve a desired position and orientation of the robot's end-effector.



Dynamics in Motion Control

- Dynamics studies the forces and torques that cause robot motion.
- Use Newton-Euler or Lagrangian methods to describe the motion.
- Stability, tracking desired trajectories, and minimizing energy consumption.

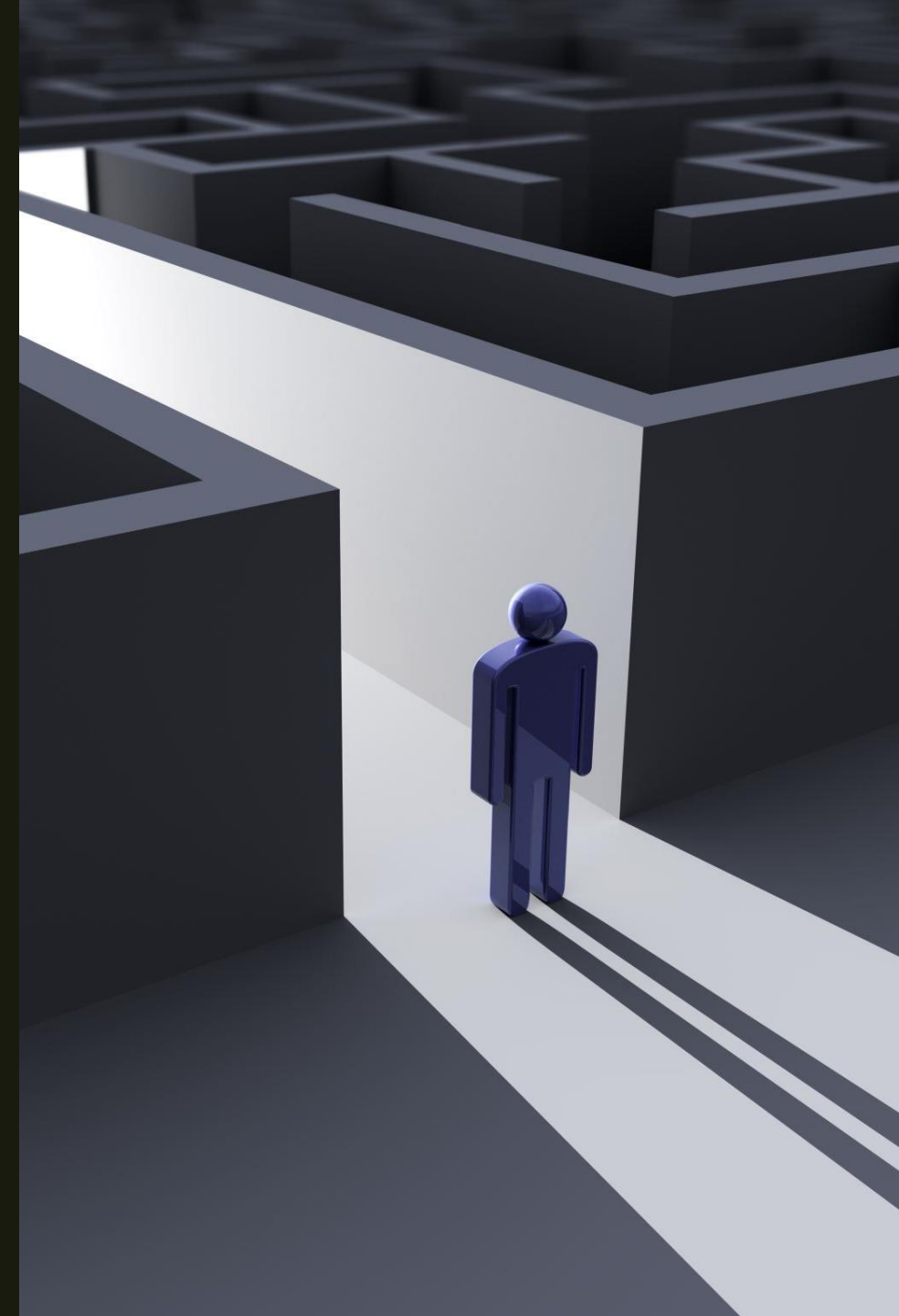


Control Strategies

- **Proportional-Integral-Derivative (PID) Control:**
 - *A control loop mechanism employing feedback to maintain the desired output.*
 - $u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$
 - *$u(t)$ is the control input, $e(t)$ is the error, K_p is the proportional gain, K_i is the integral gain, and K_d is the derivative gain.*
- **Model Predictive Control (MPC):**
 - *An advanced control strategy that uses a model of the system to predict future states and optimize control inputs.*
 - **Advantages:** *Handles multivariable control problems and constraints effectively.*

Path Planning and Navigation

- **Path Planning:**
 - *The process of determining a feasible path from the start to the goal position.*
 - **Algorithms:** *A* algorithm, Dijkstra's algorithm, Rapidly-exploring Random Trees (RRT).*
- **Navigation:**
 - *The process of guiding the robot along the planned path while avoiding obstacles.*
 - **Techniques:** *Simultaneous Localization and Mapping (SLAM), sensor fusion.*



Challenges

- **Complex Environments:** Navigating and operating in dynamic and unstructured environments.
- **Energy Efficiency:** Developing locomotion mechanisms that optimize energy consumption.
- **Robust Control:** Ensuring stability and reliability in various conditions.



Summary

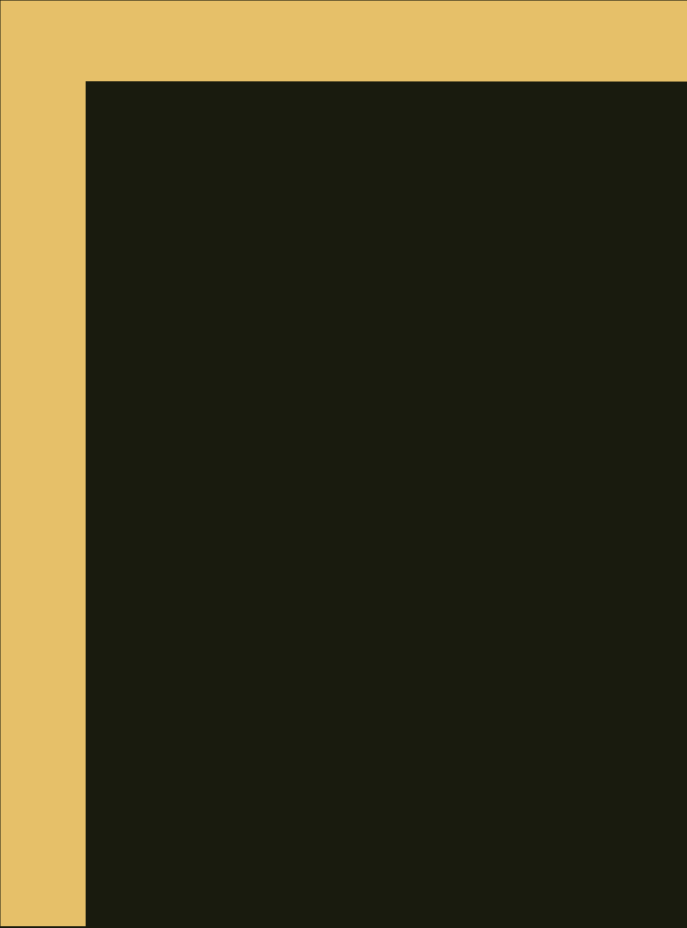
Types of Locomotion:
Wheeled, legged,
tracked, aerial, and
aquatic.

Motion Control:
Kinematics, dynamics,
control strategies, and
path planning.

Applications: Industrial,
service, healthcare,
exploration, and defense.

**Understanding
Locomotion:** Importance
of effective locomotion
mechanisms in robotics.

Control Techniques:
Various control strategies
to achieve precise and
efficient robot
movement.



Q&A